

EDUCATION

University of Guelph

Bachelor of Computing — Computer Science (Co-op), Honours

Sep 2022 – Apr 2027 (Expected)

Guelph, ON

SKILLS

Languages	Python, JavaScript / TypeScript, SQL, Java, HTML / CSS , C/C++, TypeScript, Dart
Development Tools	Git, Unix, Docker, Nginx, pgAdmin, Grafana, Metabase
Frameworks / Libraries	Django, Flask, React, SQLAlchemy, Pandas, NumPy, Selenium
Cloud / Platforms	AWS (Lambda, API Gateway, CloudWatch, Bedrock), Atlassian (Jira, Confluence)

WORK EXPERIENCE

Software Engineer Intern

TD Bank

Jan 2026 – Present

Toronto, ON

- Developing JavaScript-based automation and business logic within ServiceNow to support internal workflow automation and system integrations.
- Building data pipelines integrating internal platforms with Microsoft SQL Server to enable structured reporting and analytics.

Software Developer (Co-op)

Health Espresso Inc.

May 2025 – Aug 2025

Oakville, ON

- Shipped an editable Profile flow in Flutter using Bloc state management, with cross-field/date validation and safe rollback, and automated end-to-end tests to prevent invalid saves.
- Built Python Selenium regression suite for critical user/admin flows in the web portal using deterministic test data and explicit waits; cut manual QA by 80% and caught 15+ pre-release bugs.
- Prototyped a lightweight AWS Bedrock text-extraction service (API Gateway, Lambda, PostgreSQL) to convert free-text into patient demographics — added prompt templates, logging, and evaluation tooling while keeping latency and cost low.

Data Engineer (Co-op)

Johnson Electric

Sep 2024 – Dec 2024

Hamilton, ON

- Developed a Python-based ETL system to automate factory data processing, storing data into a PostgreSQL database with scheduled execution via Windows Task Scheduler.
- Built a Django web application to provide an interactive interface for factory operators to view, query, and manage real-time factory data, improving operational efficiency.
- Created SQL dashboards in Grafana/Metabase to visualize machine activity, downtime, and efficiency trends, used by operators and engineers to monitor production health.

Lab Software Developer (Co-op)

University of Guelph

May 2024 – Aug 2024

Guelph, ON

- Developed a Python-based EMG analysis tool with a custom GUI, capable of processing thousands of large-scale data files (up to 1,000,000 lines each), generating formatted Excel outputs, and visualizing key signal features for insights.
- Created a web-based tool to assist with transposing kinematic data from Visual3D to Excel format, allowing for easier data manipulation and analysis.

PROJECTS

PCTracker | Python, Django, MySQL, Nginx/Gunicorn | pctracker.ca

- Developed a comprehensive PC hardware platform that integrates real-time eBay pricing data, allowing users to search for instant price comparisons, build custom PC configurations with price estimates, and analyze component benchmarks.
- Utilized eBay's API with OAuth tokens to automate the extraction of real-time pricing data for second-hand computer hardware, efficiently processing up to 5000 API calls daily.
- Created a dynamic database system that stores detailed specifications and pricing for thousands of individual components, manages compatibility relationships between parts, and allows users to build and save custom PC configurations through personal accounts.
- Engineered a secure user authentication system with Django's built-in two-factor authentication, protecting user accounts and saved configurations.

Billiards Simulation | C, Python, SWIG, SQLite | github.com/jerryyou04/billiards

- Developed an interactive billiards physics simulation that accurately models ball collisions, friction, and cue interactions, allowing users to play pool games in a web browser.
- Engineered a high-performance physics engine in C that handles complex collision detection and response, calculating trajectories with mathematical precision.